

NANYANG JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATION
Higher 3

BIOLOGY

9816/01

Paper 1

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INSERT

2 hour 30 minutes

INSTRUCTIONS

You may annotate this insert and use any blank spaces for planning. **Do not write your answers** on the insert.

This document consists of **5** printed pages.

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Information for Question 1

The information provided in this insert is based on a number of published scientific articles. Other published articles may not agree with all of this information.

Circadian Clocks in Living Organisms

Scientists knew that living organisms had internal clocks centuries before they began to understand them. The body's daily rhythm is described as 'circadian', from the Latin words circa (around) and diem (day). The systems consist of a 24-hour rhythm generator – otherwise known as an oscillator.

Circadian rhythms allow living organisms to estimate the time of the day and therefore regulates the timing of physiological and cellular processes. This internal clock exists in all multi-cellular organisms, including fungi, plants, insects, and humans. It was once thought to be an exclusive feature of eukaryotes but scientists have subsequently found that single-celled cyanobacteria also display circadian rhythm.

In plants, research has revealed the involvement of the internal circadian clock in the regulation of photosynthesis. This has allowed certain plants to better conserve water and tolerate drought. Such findings are valuable and could be used to help produce new crops that can thrive in previously inhospitable, hot and dry regions across the world.

In humans, the circadian clock is synchronised to the 24-hour day by environmental inputs, most importantly by sunlight, as well as by social rhythm such as common meal times and physical exercise. It has also been linked to physiological processes such as the regulation of body temperature and hormone release.

Fig. 1.1 shows the typical events which occur during the homeostatic regulation of body temperature.

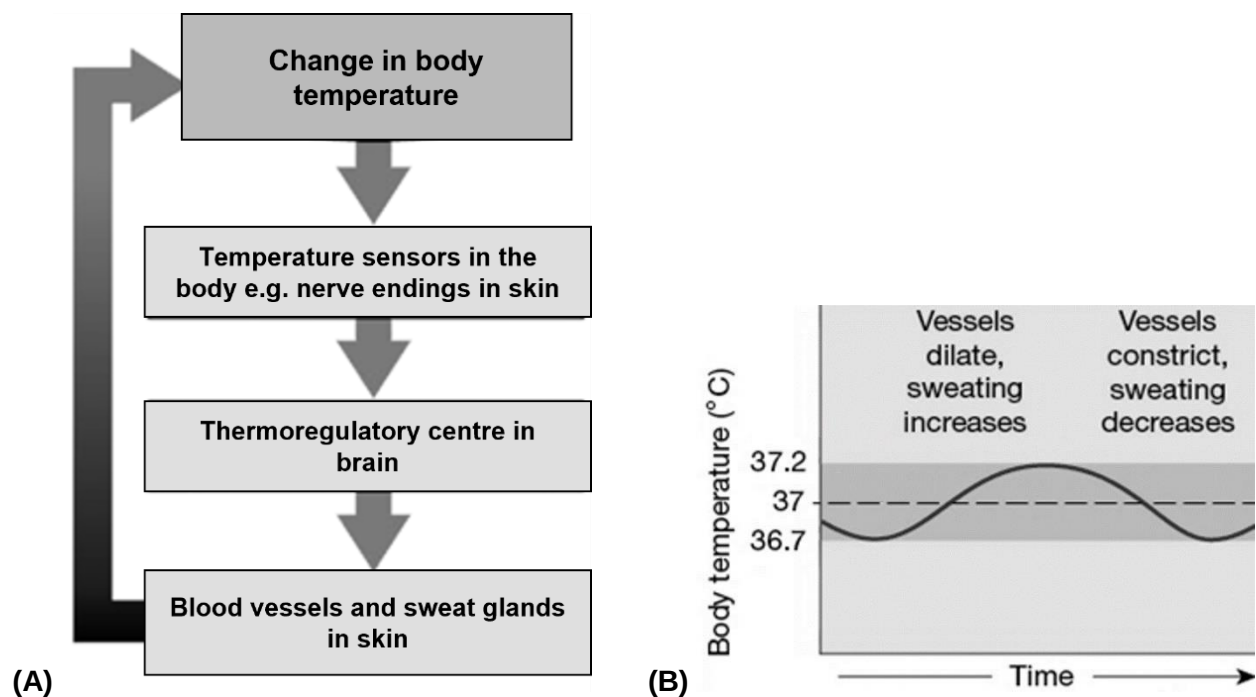


Fig. 1.1

Disruption of Circadian Clock and Disorders

Serotonin is a key hormone important for the stabilisation of mood and induces happiness through the transmission of nerve impulses.

Fig. 1.2 shows how serotonin works in the transmission of nerve impulse across the synapse.

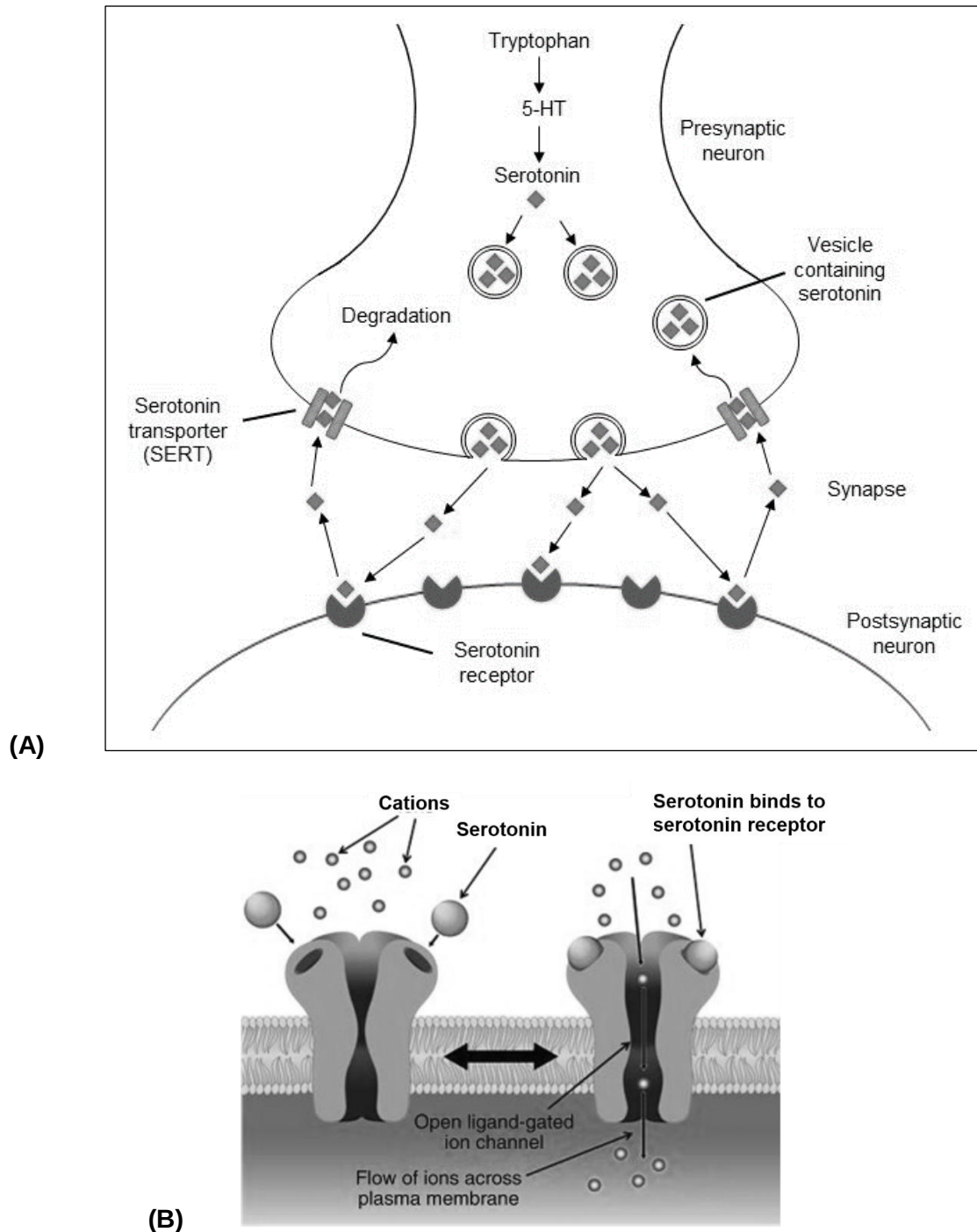


Fig. 1.2

The dysregulation of circadian rhythmicity has been associated with mood disorders such as depression.

The disruption of the circadian clock is commonly observed in neurodegenerative disorders, such as human prion diseases. In a recent study conducted, almost 90% of sporadic Creutzfeldt-Jakob disease (spCJD) patients reported sleep dysfunction during clinical evaluation, making sleep disturbance more prevalent than any other diagnostic criteria for CJD. Further research is being conducted to elucidate the role of the cellular prion protein (PrP^{C}) in circadian rhythmic regulation.

Fig. 1.3 shows the structure of a normal cellular prion protein.

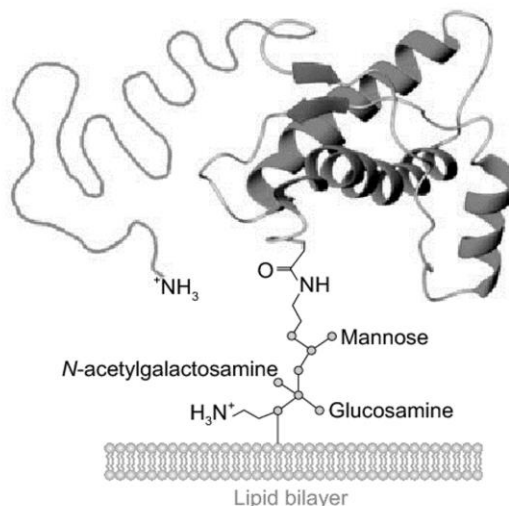


Fig. 1.3

A student investigated the number of sleep hours in two groups, control (non-CJD individuals) and CJD patients.

The number of sleep hours was measured for a night in ten individuals in the control group and twelve individuals in the CJD group. The mean age for both groups are 57.5.

Table 1.1 shows the results of the investigation.

Table 1.1

sample number	number of sleep hours / hours	
	control group	CJD
1	9	8
2	10	7
3	11	6
4	7	7
5	7	5
6	11	6
7	12	7
8	12	9
9	9	10
10	10	4
11		5
12		6
mean ($\bar{x}$)	9.80	6.67
standard deviation (s)	3.29	2.97
variance (s^2)	10.82	8.82

A t-test can be used to determine whether there is any significant difference in sleep hours between the control group and CJD group.

Table 1.2 shows the critical values for t at several different probabilities and degrees of freedom.

Table 1.2

degrees of freedom	probability, p				
	0.5	0.1	0.05	0.01	0.001
1	1.00	6.31	12.71	63.66	636.62
2	0.82	2.92	4.30	9.92	31.60
3	0.76	2.35	3.18	5.84	12.92
4	0.74	2.13	2.78	4.60	8.61
5	0.73	2.02	2.57	4.03	6.87
6	0.72	1.94	2.45	3.71	5.96
7	0.71	1.89	2.36	3.50	5.41
8	0.71	1.86	2.31	3.36	5.04
9	0.70	1.83	2.26	3.25	4.78
10	0.70	1.81	2.23	3.17	4.59
11	0.70	1.80	2.20	3.11	4.44
12	0.70	1.78	2.18	3.05	4.32
13	0.69	1.77	2.16	3.01	4.22
14	0.69	1.76	2.14	2.98	4.14
15	0.69	1.75	2.13	2.95	4.07
16	0.69	1.75	2.12	2.92	4.01
17	0.69	1.74	2.11	2.90	3.97
18	0.69	1.73	2.10	2.88	3.92
19	0.69	1.73	2.09	2.86	3.88
20	0.69	1.72	2.09	2.85	3.85